

Native Time–Energy Reciprocity

Replacing a borrowed justification (Doc. 306)

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Time increment (abstract, steps (2) and (3)): "minimal increment $d_1 = 1/75$, no time increment is smaller" → the increments $d_n = 100 \xi_n$ decrease monotonically from $d_1 = 1/75$ ($d_n \approx 1/(n + 74)$); the lower time bound is $T_0 = \xi t_P$ (time dressing of L_0 , Doc. 180, R90). The conclusion stands.

Doc. 306 Native time–energy reciprocity — replacing a borrowed justification

Abstract. Documents 025 and 063 justify the absence of a cosmic initial singularity by Heisenberg's energy–time uncertainty relation. That justification is here *replaced*, not refuted: the conclusion (no singular beginning) stands, the derivation is replaced by a native one. Three defects are named — an inverted implication (a finite Δt yields a *finite* bound, not $\Delta E \rightarrow \infty$), a misapplication (the Δt of the relation is an observable's characteristic change time, not a cosmic age), and a breach of style ("irrefutably" is not a ledger term). The replacing chain is native: from $T \cdot m = 1$ (Doc. 003/010), hence $T \cdot E = 1$, it follows that $E \rightarrow \infty \Leftrightarrow T \rightarrow 0$; but FFGFT time is bounded below by $T_0 = \xi t_P$, fixed by $\xi = 4/30000$ (Corrected on 1 Oct. 2026: previously "a minimal, non-vanishing increment $d_1 = 100 \xi_1 = 1/75$ " — the increments $d_n = 100 \xi_n$ decrease monotonically from d_1 on, d_1 is the largest; the bound is $T_0 = \xi t_P$, Doc. 180, R90.). T is therefore bounded below and E bounded above — an infinite density is structurally excluded. In addition $\partial T^4 = \emptyset$: the question of a "beginning" presupposes a boundary that a compact torus does not have. The same chain determines the black hole: the horizon is the *extremum* — slowest time $T = T_0$, maximal concentration $E = 1/T_0$, maximal curvature, all finite and well defined — no singularity, and hence nothing to repair. The same holds at the other end: $T \cdot m = 1$ has no solution at $m = 0$ — FFGFT knows no *empty* state, only a thinnest one. Both edges are excluded: the dense one is

quantified from ξ ($T_0 = \xi t_p$, $E_{\max} = E_p/\xi$, with exact closure $T_0 E_{\max} = 1$; simultaneously the natural UV cutoff of the field theory), the thin one structurally, but unquantified — the latter no deficiency but a consequence of $T \cdot m = 1$: each system carries its own upper bound $R_{\text{Torus}}(m) = \hbar/mc$, so the cosmological sector requires its own scale.

.1 What is being replaced

In their original versions, Docs. 025 and 063 both contain the passage that Heisenberg's uncertainty relation $\Delta E \cdot \Delta t \geq \hbar/2 = 1/2$ (natural units) "irrefutably proves" a classical Big-Bang singularity to be impossible, and infer:

- a temporal beginning would mean Δt finite;
- this leads to $\Delta E \rightarrow \infty$;
- hence the universe must have existed eternally.

The *conclusion* — no singular beginning, instead a finite core with a minimal scale L_0 from ξ — is retained. The *derivation* is replaced. Three reasons.

.2 Three defects of the borrowed justification

(M1) Inverted implication. The inference " Δt finite $\Rightarrow \Delta E \rightarrow \infty$ " is invalid. From $\Delta E \geq 1/(2\Delta t)$, a *finite* Δt yields a *finite* lower bound on ΔE ; a beginning a finite time ago would give a tiny, harmless bound. $\Delta E \rightarrow \infty$ follows only from $\Delta t \rightarrow 0$, i.e. from the *instant* $t = 0$, not from a finite beginning. This is not a slip but a reversed implication.

(M2) Misapplication. Energy-time uncertainty is not a canonical uncertainty relation. By Pauli's theorem there is no self-adjoint time operator conjugate to a Hamiltonian bounded below; the rigorous form (Mandelstam–Tamm) defines $\Delta t = \Delta A/|d\langle A \rangle/dt|$ — the characteristic *change time of an observable*, not an elapsed duration. The "age of the universe" is not an admissible Δt for this relation.

(M3) Breach of style. "Irrefutably proves" is a strength of assertion that does not occur in the FFGFT ledger. The ledger knows *proven* (derived from ξ), *restricted*, and *blocked*; a statement is certified as sufficiently precise to be decidable, not as irrefutable.

The overarching point. Heisenberg's relation is not false within the QM framework, but it is *projected*: it belongs to the description level that FFGFT understands as a projection from the deeper structure. An argument against the cosmic singularity that leans on it does not leave the projection — it circulates within it. This is exactly what Doc. 298 warns of: one does not exit the projection by importing its theorems.

.3 The replacing native chain

(1) Exact reciprocity. The time–mass duality (Doc. 003, cf. Doc. 010) reads

$$T \cdot m = 1, \quad \text{and with } E = m \text{ (natural units)} \quad \boxed{T \cdot E = 1} \quad (.1)$$

This is an *equation*, not an inequality — a sharper statement than $\Delta E \cdot \Delta t \geq 1/2$, and a native one.

(2) The time increments are finite. FFGFT time is the cumulated defect of the unrolled winding,

$$D(k) = \sum_{n \leq k} 100 \xi_n \rightarrow \ln\left(1 + \frac{k}{75}\right), \quad \xi_{n+1} = \xi_n (1 - 100 \xi_n), \quad (.2)$$

with the first step

$$d_1 = 100 \xi_1 = 100 \cdot \frac{4}{30000} = \frac{1}{75}. \quad (.3)$$

To be precise: $D(k)$ grows without bound as $k \rightarrow \infty$ (logarithmically). The singularity-free statement is not “ D never diverges” but: *$D(k)$ is finite at every finite k , and there is no point at which D or its increments diverge.* The first step is finite and fixed by ξ . The increments $d_n = 100 \xi_n$ decrease monotonically from $d_1 = 1/75$ ($d_2 = 0.013156$, $d_n \approx 1/(n + 74)$); the lower time bound follows from $T_0 = \xi t_p$, the time dressing of $L_0 = \xi \ell_p$ (Doc. 180, R90). (Corrected on 1 Oct. 2026: previously “no time increment is smaller than d_1 ”, heading “Time has a minimal increment” – from $\xi_{n+1} = \xi_n(1 - 100 \xi_n)$ one gets $d_1 = 0.013333 > d_2 = 0.013156 > \dots \rightarrow 0$; d_1 is the largest, not the smallest increment.)

(3) The inference. From $T \cdot E = 1$,

$$E \rightarrow \infty \iff T \rightarrow 0. \quad (.4)$$

But no state with $T = 0$ exists in FFGFT: time is bounded below by $T_0 = \xi t_p$ (Corrected on 1 Oct. 2026: previously “by a finite increment fixed by ξ ” – see (2); the bound is T_0 , not d_1). Hence E is bounded above — an infinite energy density is *structurally* excluded, not by a statement about measurement precision. The exact numerical bound depends on the normalisation and is not asserted here as a number; the structural statement stands without it.

(4) The boundary question. The torus is compact and boundaryless, $\partial T^4 = \emptyset$ (Doc. 302; cf. Doc. 263, Doc. 296). The question “what was at $t = 0$?” presupposes a boundary at which a beginning could sit. A compact torus has none. The question is therefore not wrongly *answered* but wrongly *posed*: it imports the decompactified projection as ontology. What standard cosmology reads as a singular beginning is the point at which the *projection* ends — not the world.

.4 The black hole: an extremum, not a singularity

The same chain yields the FFGFT statement about black holes — and it is unspectacular: nothing diverges.

From $T \cdot E = 1$ with the lower bound $T \geq T_0$ (the smallest possible time interval, denoted the same way in the original T0 notation; cf. $\Omega(T) = T_0/T$ in Doc. 095) it follows immediately that

$$E \leq \frac{1}{T_0}. \quad (.5)$$

The extrema are *quantified* in the corpus and derived from ξ . The minimal length $L_0 = \xi \ell_P$ (Doc. 182) yields, via energy–length reciprocity, the cutoff

$$E_{\max} = \frac{\hbar c}{L_0} = \frac{E_P}{\xi} \approx 7500 E_P \approx 9.16 \times 10^{22} \text{ GeV}, \quad T_0 = \xi t_P \approx 7.19 \times 10^{-48} \text{ s} \quad (.6)$$

(Doc. 250, Doc. 290). The reciprocity closes *exactly*: ξ cancels,

$$T_0 \cdot E_{\max} = (\xi t_P) \cdot \frac{E_P}{\xi} = t_P \cdot E_P = 1 \quad (\text{natural units}). \quad (.7)$$

This follows from the construction: $T \cdot E = 1$ holds at the extrema, not merely in the middle range.

The same cutoff appears in field theory: the fractal torus supplies a *natural UV cutoff* at E_P/ξ , which is why traditional renormalisation is not required in the full FFGFT formulation (Doc. 019, Doc. 202); the same threshold explains the rarity of magnetic monopoles (Doc. 143). The Lagrangian does not diverge because the *geometry* supplies the cutoff — the minimal time is not a regulator added afterwards, but follows from the structure.

Scales, not totals. One clarification, without which everything below would be misread: T_0 and $E_{\max}$ are *scales*, not bounds on totals. T_0 is the minimal time *increment* of the winding; $E_{\max}$ is the maximal energy *per fundamental cell*, i.e. per elementary excitation — a *cutoff*, and a cutoff always bounds a single mode, never a sum. The total energy of a composite system may be arbitrarily large: it consists of many cells, not of denser ones. A macroscopic body is not *one* winding but very many; “the T of the Earth” is not a well-posed quantity, and $\hbar/(Mc^2)$ for a stellar mass is not a state of this relation but a misapplication. $T \cdot E = 1$ and its bounds hold *per excitation*, not per aggregate.

Accordingly, the statement about the black hole concerns *concentration*, not mass: at the horizon the *local* density reaches the cell cutoff. The *total* mass remains unbounded — a larger hole has more cells, not denser ones.

The horizon is therefore the locus at which the extrema are *attained*:

- the *slowest time*: $T = T_0$, not $T \rightarrow 0$;
- the *maximal energy and mass concentration*: $E = 1/T_0$ *per cell*, finite (a density, not a total mass);
- the *maximal curvature*: finite, because it is tied to E .

Every one of these quantities is well defined. The horizon is not a boundary to something else, nor a curtain in front of a singularity, but the surface on which time attains its minimum and concentration its maximum.

One consequence deserves to be stated, because it is easily missed: in FFGFT there is *nothing to repair*. Models that replace a singular core by a bounce, a transition shell, or a daughter branch are fixing a defect that FFGFT never produces. This is not a refutation of such models — they work within a continuum picture FFGFT does not share — but the observation that the question they answer does not arise here. Where the decompactified description says “infinite”, the compact one says only: *this is the extremum*.

.5 The other extreme: the thinnest state

The reciprocity has two ends. One is the horizon; the other is usually called the “vacuum” — and precisely that name misleads.

The apparent circle. The objection runs: the vacuum is empty; it appears non-empty only through curvature; but curvature presupposes mass. No mass, no curvature; no curvature, nothing to fill the emptiness — the circle closes.

The circle is an import. It arises from a *causal arrow* that General Relativity brings with it: $T_{\mu\nu}$ generates curvature; mass is cause, curvature effect. In that picture one must ask “what is curving here?”, and without mass there is no answer. FFGFT has no such arrow. $T \cdot m = 1$ is not a cause–effect relation but a *reciprocity*: mass and temporal curvature are two readings of the same structure; neither generates the other. The question “which came first — mass or curvature?” is as wrongly posed as the question about $t = 0$: it imports a causal structure the compact description does not have. Nothing needs generating, and the circle dissolves.

Both ends are excluded. The fundamental relation itself forbids the edges:

$$m \rightarrow \infty \Rightarrow T \rightarrow 0 \quad \text{excluded by } T \geq T_0 \text{ } (L_0 = \xi \ell_p), \quad (.8)$$

$$m \rightarrow 0 \Rightarrow T \rightarrow \infty \quad \text{excluded: } T \cdot m = 1 \text{ has no solution at } m = 0. \quad (.9)$$

FFGFT therefore knows *no empty state* — not because something fills it, but because the theory contains no description of “nothing”. The torus with its winding *is* the object; “empty” would mean “no winding”, and the winding is what time is. “Vacuum” asserts an emptiness that does not exist in this framework; *the thinnest state* or *the flattest winding* are the apter terms.

The asymmetry, stated honestly. The two ends are not equally well determined, and this should not be made to look tidier than it is:

- The *dense* pole is hard and derived: T_0 and $L_0 = \xi \ell_p$ follow from ξ (Doc. 182: the chain $\xi \rightarrow m_e \rightarrow G \rightarrow \ell_p \rightarrow L_0$ depends on ξ alone).

- The *thin pole* is *structurally excluded but not quantified*. That $m = 0$ is impossible follows from the relation; *how small* m can become does not — and for a *structural* reason, not because of a gap in the derivation. Doc. 182: “The question of a universal maximal size of the universe is not well posed: every physical scale has its own system-specific upper bound $R_{\text{Torus}}(m) = \hbar/mc$.” That is $T \cdot m = 1$ in geometric form. If each mass carries its own torus radius, the cosmological scale *cannot* follow from ξ alone: it belongs to its own system and must be set independently. A separate cosmological scale is therefore *necessary*, not deficient.

A wall below, then, and no universal floor above — but to each system its own horizon.

What P35 concerns — and what it does not. Two things must be kept apart. That the cosmological sector needs its own scale is structural and unobjectionable. Writing it as $E_H = E_0 \xi^{41/4}$ is not: the exponent $41/4$ is calibrated to H_0 , not forward-derived, and the ξ -notation suggests a derivation that does not exist (P35/P36; Doc. 182 says so itself; Doc. 190). The objection concerns the *notation*, not the necessity of the separate scale.

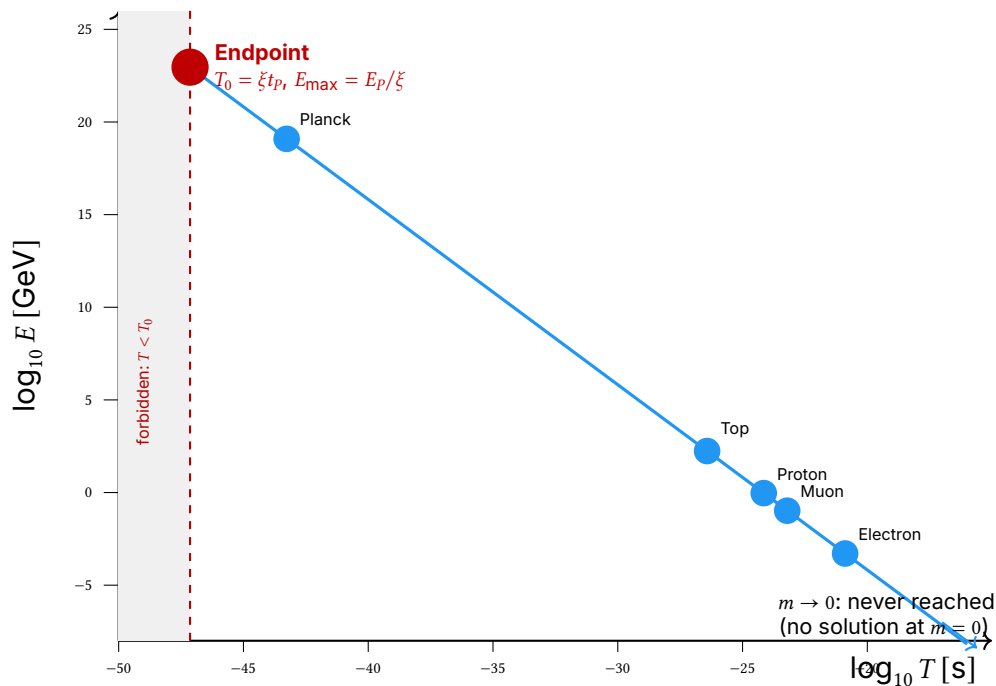


Figure 1: The relation $T \cdot E = 1$ (in SI: $T \cdot E = \hbar$) in doubly logarithmic coordinates: a *straight line* of slope -1 . All elementary excitations lie exactly on it — none is privileged, and the line has *no middle*. Exactly one point is distinguished: the **endpoint** $(T_0, E_{\max}) = (\xi t_p, E_p/\xi)$, where the curve *stops* — the wall that follows from ξ . At the other end it continues as an *asymptote* and never arrives ($T \cdot m = 1$ has no solution at $m = 0$). The two edges are therefore of different kinds: on the left an attained endpoint, on the right an unreachable limit. What is plotted are *scales* (energy per excitation), not total masses; macroscopic bodies do not belong on this axis.

And in between — a projection, not a structure. It is tempting to arrange the systems along a common axis, from maximal to minimal mass concentration, and to find oneself in the middle of it. That axis, however, is an *external view*: it presupposes a standpoint from which all systems appear side by side. FFGFT has no such standpoint. $T \cdot m = 1$ holds *per system*; there is no common T against which to plot, and no scale on which a system could locate itself.

More sharply: we *project ourselves into the middle*. The axis arises with the observer as its origin — whoever lays down the measure necessarily stands where the measure begins. The middle is therefore not a discovery but a positing, and it goes unnoticed because the one who posits and the one who is located are the same. For the same reason the edges appear “extreme”: they are what lies far from the chosen origin. *Extreme* here means *distant*, not *privileged*.

Intrinsic time is untouched by this — it is intrinsic. For an electron it is of no consequence where an outside observer enters it: its winding is the same in the laboratory as near a horizon, since $T = 1/m$ and m is the same. What differs is only how another system sees it — a relation between two, not a property of one. The “slowest time” at the horizon, too, is a statement about the comparison, not about the experience.

The projection is therefore not invalid but a *tool*: without a common axis nothing could be compared, and understanding requires comparison. It is a means of access, not a constituent of the structure. It is the same move that dissolved the question of $t = 0$ and the singularity, applied a third time — now to the scale itself. And it is a pointed form of the self-reference (Doc. 248): the thought that lays out the order is itself a process within it; it cannot stand outside, but *locates itself in the act of ordering*.

.6 An observation on the genesis

It should be noted that 025 and 063 already point to ξ in their *result*: the singularity is said to be “replaced by a tiny but finite core with minimal length scale L_0 (from ξ)”. The answer therefore already came from ξ ; only the ladder climbed to reach it was borrowed. Why the import was made although Doc. 003/010 was available is a *historical question about the genesis of the corpus*, not a systematic one. It is marked as such here and left unanswered; any answer would be speculation.

.7 Status

Correction / replacement. The conclusion (no singular beginning; a finite core with a minimal scale set by ξ) is unchanged. The justification in Docs. 025 and 063 is *no longer authoritative* for this point; it is not recorded as refuted, but as superseded by the native chain.

Proven (internal to FFGFT). $T \cdot m = 1$ (Doc. 003/010); $D(k)$ and $d_1 = 1/75$ from ξ and the recursion (Doc. 295); $\partial T^4 = \emptyset$ (Doc. 302). The inference $E \rightarrow \infty \Leftrightarrow T \rightarrow 0$ follows algebraically from the reciprocity; so does the bound $E \leq 1/T_0$ and with it the characterisation of the horizon as an extremum (no singularity). So does the exclusion of $m = 0$ (no empty state).

Quantified (proven). The extrema of the dense pole are derived from ξ and computed in the corpus: $T_0 = \xi t_P \approx 7.19 \times 10^{-48}$ s and $E_{\max} = \hbar c / L_0 = E_P / \xi \approx 9.16 \times 10^{22}$ GeV (Docs. 182, 250, 290), with the exact closure $T_0 \cdot E_{\max} = t_P \cdot E_P = 1$. The fractal torus thereby also supplies the natural UV cutoff of the field theory (Docs. 019/202/143).

Structural, not deficient. The *thin* pole remains excluded but unquantified — and that is a statement of the structure, not a gap: from $T \cdot m = 1$ it follows that every system carries its own upper bound $R_{\text{Torus}}(m) = \hbar / mc$; a universal maximal size is not meaningfully demanded (Doc. 182). The cosmological sector therefore requires its own, independently set scale.

Open (P35). Its *notation* remains to be cleaned up: $E_H = E_0 \xi^{41/4}$ calibrates the exponent to H_0 and dresses a measured quantity in ξ -form; that is not a derivation (P35/P36, Docs. 182/190). The objection concerns the notation, not the necessity of the separate scale. The connection $R_H \rightarrow \ell_1$ remains open (P20, cf. Doc. 267); the cosmological sector as a whole remains the least determined area of the corpus.